

Topic C1

States of matter

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, chemical reasoning, worked applications, common misconceptions and original exam-style practice for this topic.

Syllabus map

c1.1	Solids, liquids and gases
c1.2	Diffusion

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

1. The particle model

All matter is made of particles in constant motion. In a solid, particles are closely packed in a regular arrangement and vibrate about fixed positions. In a liquid, particles remain close but move past one another. In a gas, particles are far apart, move rapidly and randomly, and collide with the container. The model explains shape, volume, compressibility and density.

Examiner insight

Describe particles using arrangement, separation, motion and forces; do not say that particles themselves expand.

2. Changes of state

Melting and boiling require energy to overcome attractive forces between particles. Freezing and condensation release energy as stronger attractions form. Temperature remains constant during a pure substance's change of state because supplied energy changes potential energy rather than average kinetic energy.

Worked example

On a heating curve, a horizontal section at the boiling point represents liquid and gas present together while energy separates particles.

Examiner insight

Evaporation occurs only at the surface and at any temperature; boiling occurs throughout the liquid at a fixed temperature.

3. Gas pressure and temperature

Gas pressure is caused by particles colliding with container walls. At constant volume, heating increases average kinetic energy, so collisions are more frequent and involve larger momentum changes; pressure increases. At constant temperature, decreasing volume increases collision frequency and pressure.

Examiner insight

A complete pressure explanation names collisions, their frequency or force, and the resulting change in pressure.

4. Diffusion

Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration as a result of random motion. It is faster at higher temperature because particles have greater kinetic energy. Lighter gas particles diffuse faster than heavier particles.

Worked example

Ammonia, M_r 17, diffuses faster than hydrogen chloride, M_r 36.5, so the ammonium chloride ring forms nearer the HCl end.

Examiner insight

Use relative molecular mass to compare gaseous diffusion rates; do not attribute diffusion to gravity.

Exam technique

High-value habits

Use precise chemical vocabulary, balance equations, show every calculation step, include units and state symbols where required, distinguish observations from conclusions, and answer the command word directly.

Retrieval checklist

- Recall all definitions, tests and equations without notes.
- Explain one unfamiliar observation using particles, bonding or collision theory.
- Check charge, atoms, units and significant figures in every calculation.
- Complete one timed set and mark it only after finishing.

Practice Paper

Topic C1

States of matter**Instructions**

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which statement describes particles in a liquid? [1]
 - A They vibrate only at fixed positions.
 - B They are close and move past one another.
 - C They are very far apart and stationary.
 - D They form a regular lattice and cannot move.

2. Which process occurs throughout a liquid at a fixed temperature? [1]
 - A evaporation
 - B sublimation
 - C boiling
 - D diffusion

3. Why does gas pressure increase when a sealed gas is heated? [1]
 - A Particles become larger.
 - B Particles collide more often and harder.
 - C The number of particles increases.
 - D Attractive forces become stronger.

4. Which change is exothermic? [1]
 - A melting
 - B boiling
 - C sublimation
 - D freezing

5. During melting of a pure substance, its temperature [1]
 - A increases steadily
 - B decreases steadily
 - C remains constant
 - D becomes zero

6. Diffusion is the net movement of particles from [1]
 - A low pressure to high pressure only
 - B high concentration to low concentration
 - C low temperature to high temperature
 - D liquid to solid

7. Which gas diffuses fastest at the same temperature? [1]
 - A H₂
 - B O₂
 - C Cl₂
 - D CO₂

8. Which state is easiest to compress? [1]
- A solid
 - B liquid
 - C gas
 - D all equally
9. Evaporation causes cooling because [1]
- A the slowest particles escape
 - B the highest-energy particles escape
 - C particles stop moving
 - D the liquid gains energy
10. Brownian motion provides evidence that [1]
- A particles are coloured
 - B molecules are in random motion
 - C solids contain no particles
 - D all atoms have equal mass

Section B · Structured questions

11 A student heats pure ice from -10°C until steam forms.

(a) Describe the particle arrangement and motion in ice. [2]

(b) Explain why the temperature remains constant while the ice melts. [3]

(c) State two differences between evaporation and boiling. [2]

(d) Name the change from gas to liquid. [1]

[Total: 8]

12 A fixed mass of gas is heated in a rigid sealed container.

(a) State what happens to the average kinetic energy. [1]

(b) Explain, using particles, why pressure increases. [3]

(c) Explain why a gas fills its container. [2]

(d) State two assumptions of the simple particle model. [2]

[Total: 8]

13 Bromine vapour spreads through a gas jar of air.

(a) Define diffusion. [2]

(b) Explain why spreading is faster when the jar is warm. [2]

(c) State whether net movement continues after uniform colour is reached. [2]

(d) Explain why diffusion is slower in liquids than gases. [2]

[Total: 8]

14 Ammonia and hydrogen chloride gases diffuse from opposite ends of a tube.

(a) Predict where the white ring forms. [2]

(b) Explain the prediction using relative molecular masses. [3]

(c) Name the white product. [1]

(d) Write the word equation. [2]

[Total: 8]

15 A liquid is left in an open dish and its temperature falls.

(a) Name the process causing the liquid level to fall. [1]

(b) Explain the cooling effect using particle energy. [3]

(c) State two factors that increase the rate. [2]

(d) Explain why a lid reduces the rate of loss. [2]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award method and chemistry marks where shown, then identify the first point at which an answer becomes incorrect.

Solutions · Multiple choice

- 1 B. Liquid particles remain close but can change neighbours.
- 2 C. Boiling occurs throughout the liquid at its boiling point.
- 3 B. Higher kinetic energy produces more frequent, larger-force collisions.
- 4 D. Freezing releases energy as particles form stronger attractions.
- 5 C. Energy overcomes attractions rather than increasing kinetic energy.
- 6 B. That is the defining concentration gradient.
- 7 A. Hydrogen has the lowest relative molecular mass.
- 8 C. Gas particles have large spaces between them.
- 9 B. Removing the highest-energy particles lowers average kinetic energy.
- 10 B. Visible particles move irregularly due to molecular collisions.

Solutions · Structured questions

11. 11(a) Regular, closely packed particles vibrating about fixed positions [2]. (b) Energy is used to overcome attractive forces [1], increasing potential rather than kinetic energy [1], so temperature remains constant [1]. (c) Evaporation is surface-only and can occur below boiling point; boiling occurs throughout at a fixed temperature [2]. (d) Condensation [1].

12. 12(a) It increases [1]. (b) Particles move faster [1], collide more frequently / with greater momentum change [1], so force per area increases [1]. (c) Particles move rapidly and randomly with negligible attractions [2]. (d) Any two valid model assumptions, e.g. tiny particles, random motion, elastic collisions [2].

13. 13(a) Net movement from high to low concentration due to random motion [2]. (b) Higher temperature gives greater kinetic energy / faster particles [2]. (c) No net movement, but particles continue random motion [2]. (d) Liquid particles are closer and collisions/interactions hinder movement [2].

14. 14(a) Nearer the hydrogen chloride end [2]. (b) NH_3 has lower M_r [1], so ammonia particles move/diffuse faster [1] and travel farther [1]. (c) Ammonium chloride [1]. (d) ammonia + hydrogen chloride $\rightarrow$ ammonium chloride [2].

15. 15(a) Evaporation [1]. (b) Highest-energy particles escape [1], lowering average kinetic energy [1] and temperature [1]. (c) Any two: higher temperature, larger surface area, greater air movement, lower humidity [2]. (d) Vapour accumulates above liquid / increases condensation, reducing net escape [2].

Self-assessment

Circle the first incorrect step and classify it as knowledge, equation or formula, mole ratio, unit conversion, observation, practical method or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.